

# AMR Simulation Performance Report

Source CSV: [dynamic\\_tasks.csv](#)

This report summarises task completion, wait time, lift usage, and resource recommendations derived from the simulation event log.

## Executive summary

| metric                                  | value               |
|-----------------------------------------|---------------------|
| Simulation start                        | 01/01/2026 08:00:00 |
| Simulation finish                       | 02/01/2026 08:13:30 |
| Simulation duration                     | 24:13:30            |
| AMRs observed                           | 2                   |
| Tasks total                             | 12                  |
| Tasks completed                         | 12                  |
| Tasks failed                            | 0                   |
| AMR routes observed                     | 3                   |
| Total AMR route time                    | 35:26               |
| AMR workload model requirement          | 1                   |
| AMR route concurrency requirement       | 2                   |
| Total waiting time                      | 0:00                |
| Total lift time                         | 2:51                |
| Average lift utilisation                | 0.1%                |
| Recommended AMRs                        | 2                   |
| Recommended lifts                       | 1                   |
| Deliveries blocked by location capacity | 0                   |
| Locations with capacity failures        | 0                   |
| Additional inventory spaces required    | 0                   |
| Additional storage area required        | 0.00 m <sup>2</sup> |

## Method

| item              | detail                                                                                                                                                                                          |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Recommended AMRs  | Maximum of actual AMR route-time workload and 95th percentile AMR route concurrency using target utilisation 85%. Multi-stop payload tasks sharing one route are counted once for fleet demand. |
| Recommended lifts | Maximum of lift occupancy model and 95th percentile concurrent lift demand using target utilisation 70%.                                                                                        |
| Lift parsing      | Uses segment_type = lift_transfer and parses lift/floor from from_location and to_location.                                                                                                     |
| Idle percentage   | Calculated against the full simulation duration for each AMR and each lift.                                                                                                                     |

AMR list

| AMR ID | Qty | Payload (kg) | Speed (m/s) | Battery (kWh) | Charge rate (kW) | Recharge threshold (%) | Start SoC % | Start location |
|--------|-----|--------------|-------------|---------------|------------------|------------------------|-------------|----------------|
| AMR-A  | 1   | 5.0          | 1.2         | 10.0          | 3.0              | 20.0                   | 100.0       | AMR-CENTRE     |
| AMR-B  | 1   | 500.0        | 1.2         | 10.0          | 3.0              | 20.0                   | 100.0       | AMR-CENTRE     |

## AMR fleet summary

All times in the format of mm:ss

| AMR ID  | Tasks | Completed | Failed | Routes | Route time | Wait time | Avg task | Distance (km) | Recharges | Energy kWh |
|---------|-------|-----------|--------|--------|------------|-----------|----------|---------------|-----------|------------|
| AMR-A-1 | 10    | 10        | 0      | 2      | 26:56      | 0:00      | 9:00     | 0.67          | 0         | 0.0        |
| AMR-B-1 | 2     | 2         | 0      | 1      | 8:30       | 0:00      | 4:15     | 0.48          | 0         | 0.0        |
| Total   | 12    | 12        | 0      | 3      | 35:26      | 0:00      | 6:37     | 1.15          | 0         | 0.0        |

## AMR Utilisation, Idle and Wait %

| AMR ID  | Routes | Util % | Idle % | Wait % |
|---------|--------|--------|--------|--------|
| AMR-A-1 | 2      | 1.9    | 98.1   | 0.0    |
| AMR-B-1 | 1      | 0.6    | 99.4   | 0.0    |

## AMR recharge summary

No AMR recharge events were identified in the CSV.

## Lift usage summary

All times in the format of mm:ss

## Lift energy consumption

| Lift   | Trips | Total lift time | Avg trip | kWh Consumed | Util % | Idle % |
|--------|-------|-----------------|----------|--------------|--------|--------|
| Lift-2 | 4     | 0:57            | 0:14     | 0.0          | 0.1    | 99.9   |
| Lift-7 | 7     | 1:53            | 0:16     | 0.125        | 0.1    | 99.9   |
| Total  | 11    | 2:51            | 0:15     | 0.125        |        |        |

## Lift wait schedule

No lift wait events were identified in the CSV.

## Payload schedule

| Payload      | No. transported | Payload kg |
|--------------|-----------------|------------|
| Pharmacy Box | 8               | 4.0        |
| food_trolley | 1               | 120.0      |

## Location space utilisation

Locations are mapped to departments and service categories from the simulator JSON. Area is calculated from each location bounding box; utilisation compares defined inventory-space area with the location area where inventory spaces are present.

| Department | Category | Location               | Floor | Length m | Width m | Area m <sup>2</sup> | Spaces | Completed | Failed | Capacity failures | Util % | Recommended area m <sup>2</sup> | Recommended spaces |
|------------|----------|------------------------|-------|----------|---------|---------------------|--------|-----------|--------|-------------------|--------|---------------------------------|--------------------|
| -          | -        | ADULTIP1-DELIVERY AREA | 3     | 0.0      | 0.0     | 0.0                 | 0      | 0         | 0      | 0                 | 0.0    | 0.0                             | 0                  |
| -          | -        | ADULTIP1-DELIVERY HUB  | 3     | 0.0      | 0.0     | 0.0                 | 0      | 0         | 0      | 0                 | 0.0    | 0.0                             | 0                  |
| -          | -        | ADULTIP1-DH            | 3     | 0.0      | 0.0     | 0.0                 | 0      | 0         | 0      | 0                 | 0.0    | 0.0                             | 0                  |
| -          | -        | ADULTIP2-DELIVERY AREA | 3     | 0.0      | 0.0     | 0.0                 | 0      | 0         | 0      | 0                 | 0.0    | 0.0                             | 0                  |
| -          | -        | ADULTIP2-DELIVERY HUB  | 3     | 0.0      | 0.0     | 0.0                 | 0      | 0         | 0      | 0                 | 0.0    | 0.0                             | 0                  |
| -          | -        | ADULTIP2-DH            | 3     | 0.0      | 0.0     | 0.0                 | 0      | 0         | 0      | 0                 | 0.0    | 0.0                             | 0                  |
| -          | -        | ADULTIP3-DELIVERY AREA | 3     | 0.0      | 0.0     | 0.0                 | 0      | 0         | 0      | 0                 | 0.0    | 0.0                             | 0                  |
| -          | -        | ADULTIP3-DELIVERY HUB  | 3     | 0.0      | 0.0     | 0.0                 | 0      | 0         | 0      | 0                 | 0.0    | 0.0                             | 0                  |
| -          | -        | ADULTIP3-DH            | 3     | 0.0      | 0.0     | 0.0                 | 0      | 0         | 0      | 0                 | 0.0    | 0.0                             | 0                  |
| -          | -        | ADULTIP4-DELIVERY AREA | 3     | 0.0      | 0.0     | 0.0                 | 0      | 0         | 0      | 0                 | 0.0    | 0.0                             | 0                  |
| -          | -        | ADULTIP4-DELIVERY HUB  | 3     | 0.0      | 0.0     | 0.0                 | 0      | 0         | 0      | 0                 | 0.0    | 0.0                             | 0                  |
| -          | -        | ADULTIP4-DH            | 3     | 0.0      | 0.0     | 0.0                 | 0      | 0         | 0      | 0                 | 0.0    | 0.0                             | 0                  |
| -          | -        | ADULTIP5-DELIVERY AREA | 4     | 0.0      | 0.0     | 0.0                 | 0      | 0         | 0      | 0                 | 0.0    | 0.0                             | 0                  |
| -          | -        | ADULTIP5-DELIVERY HUB  | 4     | 0.0      | 0.0     | 0.0                 | 0      | 0         | 0      | 0                 | 0.0    | 0.0                             | 0                  |
| -          | -        | ADULTIP5-DH            | 4     | 0.0      | 0.0     | 0.0                 | 0      | 0         | 0      | 0                 | 0.0    | 0.0                             | 0                  |



| Department | Category | Location               | Floor | Length m | Width m | Area m <sup>2</sup> | Spaces | Completed | Failed | Capacity failures | Util % | Recommended area m <sup>2</sup> | Recommended spaces |
|------------|----------|------------------------|-------|----------|---------|---------------------|--------|-----------|--------|-------------------|--------|---------------------------------|--------------------|
| -          | -        | ADULTIP6-DELIVERY AREA | 4     | 0.0      | 0.0     | 0.0                 | 0      | 0         | 0      | 0                 | 0.0    | 0.0                             | 0                  |
| -          | -        | ADULTIP6-DELIVERY HUB  | 4     | 0.0      | 0.0     | 0.0                 | 0      | 0         | 0      | 0                 | 0.0    | 0.0                             | 0                  |
| -          | -        | ADULTIP6-DH            | 4     | 0.0      | 0.0     | 0.0                 | 0      | 0         | 0      | 0                 | 0.0    | 0.0                             | 0                  |
| -          | -        | ADULTIP7-DELIVERY AREA | 4     | 0.0      | 0.0     | 0.0                 | 0      | 0         | 0      | 0                 | 0.0    | 0.0                             | 0                  |
| -          | -        | ADULTIP7-DELIVERY HUB  | 4     | 0.0      | 0.0     | 0.0                 | 0      | 0         | 0      | 0                 | 0.0    | 0.0                             | 0                  |
| -          | -        | ADULTIP7-DH            | 4     | 0.0      | 0.0     | 0.0                 | 0      | 0         | 0      | 0                 | 0.0    | 0.0                             | 0                  |
| -          | -        | ADULTIP8-DELIVERY AREA | 5     | 0.0      | 0.0     | 0.0                 | 0      | 0         | 0      | 0                 | 0.0    | 0.0                             | 0                  |
| -          | -        | ADULTIP8-DELIVERY HUB  | 5     | 0.0      | 0.0     | 0.0                 | 0      | 0         | 0      | 0                 | 0.0    | 0.0                             | 0                  |
| -          | -        | ADULTIP8-DH            | 5     | 0.0      | 0.0     | 0.0                 | 0      | 0         | 0      | 0                 | 0.0    | 0.0                             | 0                  |
| -          | -        | ADULTIP9-DELIVERY AREA | 5     | 0.0      | 0.0     | 0.0                 | 0      | 0         | 0      | 0                 | 0.0    | 0.0                             | 0                  |
| -          | -        | ADULTIP9-DELIVERY HUB  | 5     | 0.0      | 0.0     | 0.0                 | 0      | 0         | 0      | 0                 | 0.0    | 0.0                             | 0                  |
| -          | -        | ADULTIP9-DH            | 5     | 0.0      | 0.0     | 0.0                 | 0      | 0         | 0      | 0                 | 0.0    | 0.0                             | 0                  |
| -          | -        | AMR-CENTRE             | 2     | 0.0      | 0.0     | 0.0                 | 0      | 3         | 0      | 0                 | 0.0    | 0.0                             | 0                  |
| -          | -        | BIN YARD               | 0     | 0.0      | 0.0     | 0.0                 | 0      | 0         | 0      | 0                 | 0.0    | 0.0                             | 0                  |
| -          | -        | CARDIO-DELIVERY AREA   | 4     | 0.0      | 0.0     | 0.0                 | 0      | 0         | 0      | 0                 | 0.0    | 0.0                             | 0                  |
| -          | -        | CARDIO-DELIVERY HUB    | 4     | 0.0      | 0.0     | 0.0                 | 0      | 0         | 0      | 0                 | 0.0    | 0.0                             | 0                  |
| -          | -        | CARDIO-DH              | 4     | 0.0      | 0.0     | 0.0                 | 0      | 0         | 0      | 0                 | 0.0    | 0.0                             | 0                  |

| Department | Category | Location                   | Floor | Length m | Width m | Area m <sup>2</sup> | Spaces | Completed | Failed | Capacity failures | Util % | Recommended area m <sup>2</sup> | Recommended spaces |
|------------|----------|----------------------------|-------|----------|---------|---------------------|--------|-----------|--------|-------------------|--------|---------------------------------|--------------------|
| -          | -        | ELECTIVE-WARD-DELIVERYAREA | 4     | 0.0      | 0.0     | 0.0                 | 0      | 0         | 0      | 0                 | 0.0    | 0.0                             | 0                  |
| -          | -        | ELECTIVE-WARD-DELIVERYHUB  | 4     | 0.0      | 0.0     | 0.0                 | 0      | 0         | 0      | 0                 | 0.0    | 0.0                             | 0                  |
| -          | -        | ELECTIVE-WARD-DH           | 4     | 0.0      | 0.0     | 0.0                 | 0      | 0         | 0      | 0                 | 0.0    | 0.0                             | 0                  |
| -          | -        | ENDOSCOPY-DELIVERYAREA     | 1     | 0.0      | 0.0     | 0.0                 | 0      | 0         | 0      | 0                 | 0.0    | 0.0                             | 0                  |
| -          | -        | ENDOSCOPY-DH               | 1     | 0.0      | 0.0     | 0.0                 | 0      | 0         | 0      | 0                 | 0.0    | 0.0                             | 0                  |
| -          | -        | ICU-DELIVERYHUB            | 1     | 0.0      | 0.0     | 0.0                 | 0      | 0         | 0      | 0                 | 0.0    | 0.0                             | 0                  |
| -          | -        | ICU-DH                     | 1     | 0.0      | 0.0     | 0.0                 | 0      | 0         | 0      | 0                 | 0.0    | 0.0                             | 0                  |
| -          | -        | KITCHEN                    | 2     | 0.0      | 0.0     | 0.0                 | 0      | 0         | 0      | 0                 | 0.0    | 0.0                             | 0                  |
| -          | -        | KITCHEN-DH                 | 2     | 0.0      | 0.0     | 0.0                 | 0      | 0         | 0      | 0                 | 0.0    | 0.0                             | 0                  |
| -          | -        | KITCHEN-DH2                | 2     | 0.0      | 0.0     | 0.0                 | 0      | 0         | 0      | 0                 | 0.0    | 0.0                             | 0                  |
| -          | -        | Linen Deliveries           | 0     | 0.0      | 0.0     | 0.0                 | 0      | 0         | 0      | 0                 | 0.0    | 0.0                             | 0                  |
| -          | -        | MATERNITY-DELIVERYAREA     | 3     | 0.0      | 0.0     | 0.0                 | 0      | 0         | 0      | 0                 | 0.0    | 0.0                             | 0                  |
| -          | -        | MATERNITY-DELIVERYHUB      | 3     | 0.0      | 0.0     | 0.0                 | 0      | 0         | 0      | 0                 | 0.0    | 0.0                             | 0                  |
| -          | -        | MATERNITY-DH               | 3     | 0.0      | 0.0     | 0.0                 | 0      | 0         | 0      | 0                 | 0.0    | 0.0                             | 0                  |
| -          | -        | OPD3-DELIVERYAREA          | 1     | 0.0      | 0.0     | 0.0                 | 0      | 0         | 0      | 0                 | 0.0    | 0.0                             | 0                  |
| -          | -        | OPD3-DH                    | 1     | 0.0      | 0.0     | 0.0                 | 0      | 0         | 0      | 0                 | 0.0    | 0.0                             | 0                  |
| -          | -        | OPD4-DELIVERYAREA          | 1     | 0.0      | 0.0     | 0.0                 | 0      | 0         | 0      | 0                 | 0.0    | 0.0                             | 0                  |
| -          | -        | OPD4-DH                    | 1     | 0.0      | 0.0     | 0.0                 | 0      | 0         | 0      | 0                 | 0.0    | 0.0                             | 0                  |

| Department      | Category | Location                 | Floor | Length m | Width m | Area m <sup>2</sup> | Spaces | Completed | Failed | Capacity failures | Util % | Recommended area m <sup>2</sup> | Recommended spaces |
|-----------------|----------|--------------------------|-------|----------|---------|---------------------|--------|-----------|--------|-------------------|--------|---------------------------------|--------------------|
| -               | -        | PAEDIATRIC-DELIVERYAREA  | 3     | 0.0      | 0.0     | 0.0                 | 0      | 0         | 0      | 0                 | 0.0    | 0.0                             | 0                  |
| -               | -        | PAEDIATRIC-DELIVERYHUB   | 3     | 0.0      | 0.0     | 0.0                 | 0      | 0         | 0      | 0                 | 0.0    | 0.0                             | 0                  |
| -               | -        | PAEDIATRIC-DH            | 3     | 0.0      | 0.0     | 0.0                 | 0      | 0         | 0      | 0                 | 0.0    | 0.0                             | 0                  |
| -               | -        | RD                       | 0     | 0.0      | 0.0     | 0.0                 | 0      | 0         | 0      | 0                 | 0.0    | 0.0                             | 0                  |
| -               | -        | RD2                      | 0     | 0.0      | 0.0     | 0.0                 | 0      | 0         | 0      | 0                 | 0.0    | 0.0                             | 0                  |
| -               | -        | RESPIRATORY-DELIVERYAREA | 4     | 0.0      | 0.0     | 0.0                 | 0      | 0         | 0      | 0                 | 0.0    | 0.0                             | 0                  |
| -               | -        | RESPIRATORY-DELIVERYHUB  | 4     | 0.0      | 0.0     | 0.0                 | 0      | 0         | 0      | 0                 | 0.0    | 0.0                             | 0                  |
| -               | -        | RESPIRATORY-DH           | 4     | 0.0      | 0.0     | 0.0                 | 0      | 0         | 0      | 0                 | 0.0    | 0.0                             | 0                  |
| -               | -        | STROKE-DELIVERYAREA      | 5     | 0.0      | 0.0     | 0.0                 | 0      | 0         | 0      | 0                 | 0.0    | 0.0                             | 0                  |
| -               | -        | STROKE-DELIVERYHUB       | 5     | 0.0      | 0.0     | 0.0                 | 0      | 0         | 0      | 0                 | 0.0    | 0.0                             | 0                  |
| -               | -        | STROKE-DH                | 5     | 0.0      | 0.0     | 0.0                 | 0      | 0         | 0      | 0                 | 0.0    | 0.0                             | 0                  |
| -               | -        | THEATRES-STORE           | 1     | 0.0      | 0.0     | 0.0                 | 0      | 0         | 0      | 0                 | 0.0    | 0.0                             | 0                  |
| -               | -        | THEATRES-STORE2          | 1     | 0.0      | 0.0     | 0.0                 | 0      | 0         | 0      | 0                 | 0.0    | 0.0                             | 0                  |
| -               | -        | THEATRES1-DH             | 1     | 0.0      | 0.0     | 0.0                 | 0      | 0         | 0      | 0                 | 0.0    | 0.0                             | 0                  |
| -               | -        | THEATRES2-DH             | 1     | 0.0      | 0.0     | 0.0                 | 0      | 0         | 0      | 0                 | 0.0    | 0.0                             | 0                  |
| -               | -        | UEC-DELIVERYHUB          | 0     | 0.0      | 0.0     | 0.0                 | 0      | 0         | 0      | 0                 | 0.0    | 0.0                             | 0                  |
| Ambulatory Care | Waste    | D17-WASTE                | 1     | 4.12     | 2.92    | 11.96               | 0      | 0         | 0      | 0                 | 0.0    | 11.96                           | 0                  |
| Birthing Suite  | Pharmacy | D19-PHARMACY             | 1     | 2.02     | 6.03    | 10.15               | 0      | 0         | 0      | 0                 | 0.0    | 10.15                           | 0                  |
| Birthing Suite  | Stores   | D19-STORES               | 1     | 7.19     | 6.05    | 33.44               | 0      | 0         | 0      | 0                 | 0.0    | 33.44                           | 0                  |

| Department             | Category | Location     | Floor | Length m | Width m | Area m <sup>2</sup> | Spaces | Completed | Failed | Capacity failures | Util % | Recommended area m <sup>2</sup> | Recommended spaces |
|------------------------|----------|--------------|-------|----------|---------|---------------------|--------|-----------|--------|-------------------|--------|---------------------------------|--------------------|
| Birthing Suite         | Waste    | D19-WASTE    | 1     | 4.12     | 2.93    | 11.92               | 0      | 0         | 0      | 0                 | 0.0    | 11.92                           | 0                  |
| Central Imaging        | Linen    | D8-LINEN     | 0     | 1.12     | 3.58    | 4.01                | 0      | 0         | 0      | 0                 | 0.0    | 4.01                            | 0                  |
| Central Imaging        | Linen    | D8-LINEN_2   | 0     | 1.38     | 2.45    | 3.28                | 0      | 0         | 0      | 0                 | 0.0    | 3.28                            | 0                  |
| Central Imaging        | Pharmacy | D8-PHARMACY  | 0     | 2.88     | 3.07    | 8.62                | 0      | 0         | 0      | 0                 | 0.0    | 8.62                            | 0                  |
| Central Imaging        | Stores   | D8-STORES    | 0     | 3.2      | 5.32    | 16.6                | 0      | 0         | 0      | 0                 | 0.0    | 16.6                            | 0                  |
| Central Imaging        | Waste    | D8-WASTE     | 0     | 5.54     | 5.09    | 27.71               | 0      | 0         | 0      | 0                 | 0.0    | 27.71                           | 0                  |
| ED-Paediatrics         | Pharmacy | D13-PHARMACY | 0     | 0.0      | 0.0     | 0.0                 | 0      | 0         | 0      | 0                 | 0.0    | 0.0                             | 0                  |
| ED-Paediatrics         | Stores   | D13-STORES   | 0     | 0.0      | 0.0     | 0.0                 | 0      | 0         | 0      | 0                 | 0.0    | 0.0                             | 0                  |
| Essential Services Lab | Waste    | D15-WASTE    | 0     | 5.62     | 4.14    | 23.11               | 0      | 0         | 0      | 0                 | 0.0    | 23.11                           | 0                  |
| Fraility               | Catering | D1-CATERING  | 0     | 2.81     | 1.32    | 3.61                | 2      | 1         | 0      | 0                 | 53.2   | 3.61                            | 2                  |
| Fraility               | Linen    | D1-LINEN     | 0     | 1.24     | 2.45    | 3.01                | 1      | 0         | 0      | 0                 | 29.9   | 3.01                            | 1                  |
| Fraility               | Linen    | D1-LINEN_2   | 0     | 1.77     | 1.7     | 2.99                | 1      | 0         | 0      | 0                 | 30.1   | 2.99                            | 1                  |
| Fraility               | Pharmacy | D1-PHARMACY  | 0     | 5.33     | 2.26    | 11.97               | 0      | 2         | 0      | 0                 | 0.0    | 11.97                           | 0                  |
| Fraility               | Stores   | D1-STORES    | 0     | 4.13     | 2.12    | 8.68                | 4      | 0         | 0      | 0                 | 31.3   | 8.68                            | 4                  |
| Fraility               | Waste    | D1-WASTE     | 0     | 4.15     | 3.28    | 13.45               | 4      | 0         | 0      | 0                 | 28.6   | 13.45                           | 4                  |
| Minor Injuries         | Linen    | D10-LINEN    | 0     | 1.92     | 1.58    | 2.95                | 0      | 0         | 0      | 0                 | 0.0    | 2.95                            | 0                  |
| Minor Injuries         | Pharmacy | D10-PHARMACY | 0     | 3.12     | 2.65    | 8.0                 | 0      | 0         | 0      | 0                 | 0.0    | 8.0                             | 0                  |
| Minor Injuries         | Stores   | D10-STORES   | 0     | 3.41     | 4.7     | 14.85               | 5      | 0         | 0      | 0                 | 22.9   | 14.85                           | 5                  |
| Mortuary               | Stores   | D6-STORES    | 0     | 8.83     | 4.29    | 26.45               | 0      | 0         | 0      | 0                 | 0.0    | 26.45                           | 0                  |

| Department  | Category | Location     | Floor | Length m | Width m | Area m² | Spaces | Completed | Failed | Capacity failures | Util % | Recommended area m² | Recommended spaces |
|-------------|----------|--------------|-------|----------|---------|---------|--------|-----------|--------|-------------------|--------|---------------------|--------------------|
| Mortuary    | Waste    | D6-WASTE     | 0     | 4.28     | 3.36    | 13.14   | 0      | 0         | 0      | 0                 | 0.0    | 13.14               | 0                  |
| Neonatal    | Linen    | D18-LINEN    | 1     | 3.32     | 2.98    | 9.84    | 0      | 0         | 0      | 0                 | 0.0    | 9.84                | 0                  |
| Neonatal    | Pharmacy | D18-PHARMACY | 1     | 2.88     | 5.96    | 14.46   | 0      | 0         | 0      | 0                 | 0.0    | 14.46               | 0                  |
| Neonatal    | Waste    | D18-WASTE    | 1     | 4.14     | 3.55    | 14.64   | 0      | 0         | 0      | 0                 | 0.0    | 14.64               | 0                  |
| OPD1        | Pharmacy | D4-PHARMACY  | 0     | 4.2      | 3.56    | 14.73   | 0      | 2         | 0      | 0                 | 0.0    | 14.73               | 0                  |
| OPD1        | Waste    | D4-WASTE     | 0     | 2.74     | 4.16    | 11.15   | 0      | 0         | 0      | 0                 | 0.0    | 11.15               | 0                  |
| OPD2        | Pharmacy | D5-PHARMACY  | 0     | 4.96     | 2.46    | 11.78   | 0      | 0         | 0      | 0                 | 0.0    | 11.78               | 0                  |
| OPD2        | Stores   | D5-STORES    | 0     | 4.94     | 2.13    | 10.33   | 0      | 0         | 0      | 0                 | 0.0    | 10.33               | 0                  |
| OPD2        | Waste    | D5-WASTE     | 0     | 2.83     | 4.08    | 11.38   | 0      | 0         | 0      | 0                 | 0.0    | 11.38               | 0                  |
| OPD3        | Pharmacy | D44-PHARMACY | 0     | 4.94     | 2.48    | 12.04   | 0      | 0         | 0      | 0                 | 0.0    | 12.04               | 0                  |
| OPD3        | Waste    | D44-WASTE    | 0     | 4.48     | 2.02    | 8.77    | 0      | 0         | 0      | 0                 | 0.0    | 8.77                | 0                  |
| OPD4        | Pharmacy | D45-PHARMACY | 0     | 4.97     | 2.48    | 11.95   | 0      | 0         | 0      | 0                 | 0.0    | 11.95               | 0                  |
| OPD4        | Stores   | D45-STORES   | 0     | 4.92     | 2.14    | 10.39   | 0      | 0         | 0      | 0                 | 0.0    | 10.39               | 0                  |
| OPD4        | Waste    | D45-WASTE    | 0     | 4.4      | 2.0     | 8.59    | 0      | 0         | 0      | 0                 | 0.0    | 8.59                | 0                  |
| Pharmacy    | Pharmacy | D7-PHARMACY  | 0     | 0.0      | 0.0     | 0.0     | 0      | 0         | 0      | 0                 | 0.0    | 0.0                 | 0                  |
| Pharmacy    | Stores   | D7-STORES    | 0     | 7.22     | 6.85    | 48.96   | 0      | 0         | 0      | 0                 | 0.0    | 48.96               | 0                  |
| Pharmacy    | Waste    | D7-WASTE     | 0     | 6.78     | 4.11    | 20.62   | 0      | 0         | 0      | 0                 | 0.0    | 20.62               | 0                  |
| Resturant   | Stores   | D9-STORES    | 0     | 5.55     | 5.46    | 28.27   | 0      | 0         | 0      | 0                 | 0.0    | 28.27               | 0                  |
| Resturant   | Waste    | D9-WASTE     | 0     | 3.37     | 3.59    | 11.83   | 0      | 0         | 0      | 0                 | 0.0    | 11.83               | 0                  |
| SDEC        | Pharmacy | D2-PHARMACY  | 0     | 3.97     | 5.53    | 21.74   | 0      | 2         | 0      | 0                 | 0.0    | 21.74               | 0                  |
| SDEC        | Waste    | D2-WASTE     | 0     | 4.15     | 3.4     | 13.95   | 4      | 0         | 0      | 0                 | 27.5   | 13.95               | 4                  |
| Urgent Care | Catering | D3-CATERING  | 0     | 3.09     | 1.33    | 4.01    | 0      | 0         | 0      | 0                 | 0.0    | 4.01                | 0                  |

| Department  | Category | Location      | Floor | Length m | Width m | Area m <sup>2</sup> | Spaces | Completed | Failed | Capacity failures | Util % | Recommended area m <sup>2</sup> | Recommended spaces |
|-------------|----------|---------------|-------|----------|---------|---------------------|--------|-----------|--------|-------------------|--------|---------------------------------|--------------------|
| Urgent Care | Catering | D3-CATERING_2 | 0     | 1.78     | 2.33    | 3.94                | 0      | 0         | 0      | 0                 | 0.0    | 3.94                            | 0                  |
| Urgent Care | Linen    | D3-LINEN      | 0     | 1.23     | 2.42    | 2.91                | 0      | 0         | 0      | 0                 | 0.0    | 2.91                            | 0                  |
| Urgent Care | Linen    | D3-LINEN_2    | 0     | 1.76     | 1.73    | 2.98                | 0      | 0         | 0      | 0                 | 0.0    | 2.98                            | 0                  |
| Urgent Care | Pharmacy | D3-PHARMACY   | 0     | 5.33     | 2.2     | 11.58               | 0      | 2         | 0      | 0                 | 0.0    | 11.58                           | 0                  |
| Urgent Care | Stores   | D3-STORES     | 0     | 4.17     | 1.61    | 6.63                | 0      | 0         | 0      | 0                 | 0.0    | 6.63                            | 0                  |
| Urgent Care | Waste    | D3-WASTE      | 0     | 4.14     | 3.68    | 15.03               | 0      | 0         | 0      | 0                 | 0.0    | 15.03                           | 0                  |

## Failed delivery analysis

Failed deliveries are grouped with their logged reason and payload dimensions so capacity failures can be traced back to the attempted load size.

No failed deliveries were identified in the simulation event log.

## Location storage recommendations

Recommended area keeps the current location area as a baseline and adds failed payload footprints with a 30% handling allowance. Recommended spaces add capacity-related failed attempts to the current defined inventory-space count.

| Department | Category | Location        | Current area m <sup>2</sup> | Recommended area m <sup>2</sup> | Additional area m <sup>2</sup> | Current spaces | Recommended spaces | Additional spaces | Reason                                  |
|------------|----------|-----------------|-----------------------------|---------------------------------|--------------------------------|----------------|--------------------|-------------------|-----------------------------------------|
| -          | -        | RD2             | 0.0                         | 0.0                             | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| -          | -        | THEATRES-STORE2 | 0.0                         | 0.0                             | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| -          | -        | THEATRES-STORE  | 0.0                         | 0.0                             | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| -          | -        | THEATRES1-DH    | 0.0                         | 0.0                             | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| -          | -        | THEATRES2-DH    | 0.0                         | 0.0                             | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| -          | -        | ICU-DH          | 0.0                         | 0.0                             | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| -          | -        | ENDOSCOPY-DH    | 0.0                         | 0.0                             | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| -          | -        | OPD4-DH         | 0.0                         | 0.0                             | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| -          | -        | OPD3-DH         | 0.0                         | 0.0                             | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| -          | -        | KITCHEN         | 0.0                         | 0.0                             | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| -          | -        | KITCHEN-DH      | 0.0                         | 0.0                             | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| -          | -        | AMR-CENTRE      | 0.0                         | 0.0                             | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| -          | -        | ADULTIP3-DH     | 0.0                         | 0.0                             | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |



| Department | Category | Location                  | Current area m <sup>2</sup> | Recommended area m <sup>2</sup> | Additional area m <sup>2</sup> | Current spaces | Recommended spaces | Additional spaces | Reason                                  |
|------------|----------|---------------------------|-----------------------------|---------------------------------|--------------------------------|----------------|--------------------|-------------------|-----------------------------------------|
| -          | -        | MATERNITY-DH              | 0.0                         | 0.0                             | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| -          | -        | MATERNITY-DELIVERYHUB     | 0.0                         | 0.0                             | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| -          | -        | RD                        | 0.0                         | 0.0                             | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| -          | -        | ADULTIP3-DELIVERYHUB      | 0.0                         | 0.0                             | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| -          | -        | UEC-DELIVERYHUB           | 0.0                         | 0.0                             | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| -          | -        | ENDOSCOPY-DELIVERYAREA    | 0.0                         | 0.0                             | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| -          | -        | OPD4-DELIVERYAREA         | 0.0                         | 0.0                             | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| -          | -        | OPD3-DELIVERYAREA         | 0.0                         | 0.0                             | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| -          | -        | ICU-DELIVERYHUB           | 0.0                         | 0.0                             | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| -          | -        | KITCHEN-DH2               | 0.0                         | 0.0                             | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| -          | -        | ELECTIVE-WARD-DH          | 0.0                         | 0.0                             | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| -          | -        | ELECTIVE-WARD-DELIVERYHUB | 0.0                         | 0.0                             | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| -          | -        | PAEDIATRIC-DELIVERYHUB    | 0.0                         | 0.0                             | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| -          | -        | PAEDIATRIC-DH             | 0.0                         | 0.0                             | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| -          | -        | ADULTIP4-DH               | 0.0                         | 0.0                             | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |

| Department | Category | Location                | Current area m <sup>2</sup> | Recommended area m <sup>2</sup> | Additional area m <sup>2</sup> | Current spaces | Recommended spaces | Additional spaces | Reason                                  |
|------------|----------|-------------------------|-----------------------------|---------------------------------|--------------------------------|----------------|--------------------|-------------------|-----------------------------------------|
| -          | -        | ADULTIP4-DELIVERY HUB   | 0.0                         | 0.0                             | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| -          | -        | ADULTIP1-DH             | 0.0                         | 0.0                             | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| -          | -        | ADULTIP2-DH             | 0.0                         | 0.0                             | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| -          | -        | ADULTIP1-DELIVERY HUB   | 0.0                         | 0.0                             | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| -          | -        | ADULTIP2-DELIVERY HUB   | 0.0                         | 0.0                             | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| -          | -        | RESPIRATORY-DELIVERYHUB | 0.0                         | 0.0                             | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| -          | -        | RESPIRATORY-DH          | 0.0                         | 0.0                             | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| -          | -        | CARDIO-DELIVERYHUB      | 0.0                         | 0.0                             | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| -          | -        | CARDIO-DH               | 0.0                         | 0.0                             | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| -          | -        | ADULTIP5-DH             | 0.0                         | 0.0                             | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| -          | -        | ADULTIP5-DELIVERY HUB   | 0.0                         | 0.0                             | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| -          | -        | ADULTIP6-DH             | 0.0                         | 0.0                             | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| -          | -        | ADULTIP7-DH             | 0.0                         | 0.0                             | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| -          | -        | ADULTIP6-DELIVERY HUB   | 0.0                         | 0.0                             | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| -          | -        | ADULTIP7-DELIVERY HUB   | 0.0                         | 0.0                             | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |

| Department | Category | Location                 | Current area m <sup>2</sup> | Recommended area m <sup>2</sup> | Additional area m <sup>2</sup> | Current spaces | Recommended spaces | Additional spaces | Reason                                  |
|------------|----------|--------------------------|-----------------------------|---------------------------------|--------------------------------|----------------|--------------------|-------------------|-----------------------------------------|
| -          | -        | ADULTIP8-DH              | 0.0                         | 0.0                             | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| -          | -        | STROKE-DH                | 0.0                         | 0.0                             | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| -          | -        | STROKE-DELIVERYHUB       | 0.0                         | 0.0                             | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| -          | -        | ADULTIP8-DELIVERYHUB     | 0.0                         | 0.0                             | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| -          | -        | ADULTIP9-DH              | 0.0                         | 0.0                             | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| -          | -        | ADULTIP9-DELIVERYHUB     | 0.0                         | 0.0                             | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| -          | -        | BIN YARD                 | 0.0                         | 0.0                             | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| -          | -        | Linen Deliveries         | 0.0                         | 0.0                             | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| -          | -        | PAEDIATRIC-DELIVERYAREA  | 0.0                         | 0.0                             | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| -          | -        | MATERNITY-DELIVERYAREA   | 0.0                         | 0.0                             | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| -          | -        | ADULTIP1-DELIVERYAREA    | 0.0                         | 0.0                             | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| -          | -        | ADULTIP2-DELIVERYAREA    | 0.0                         | 0.0                             | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| -          | -        | ADULTIP3-DELIVERYAREA    | 0.0                         | 0.0                             | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| -          | -        | ADULTIP4-DELIVERYAREA    | 0.0                         | 0.0                             | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| -          | -        | RESPIRATORY-DELIVERYAREA | 0.0                         | 0.0                             | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |

| Department | Category | Location                    | Current area m² | Recommended area m² | Additional area m² | Current spaces | Recommended spaces | Additional spaces | Reason                                  |
|------------|----------|-----------------------------|-----------------|---------------------|--------------------|----------------|--------------------|-------------------|-----------------------------------------|
| -          | -        | CARDIO-DELIVERY AREA        | 0.0             | 0.0                 | 0.0                | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| -          | -        | ADULTIP5-DELIVERY AREA      | 0.0             | 0.0                 | 0.0                | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| -          | -        | ELECTIVE-WARD-DELIVERY AREA | 0.0             | 0.0                 | 0.0                | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| -          | -        | ADULTIP6-DELIVERY AREA      | 0.0             | 0.0                 | 0.0                | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| -          | -        | ADULTIP7-DELIVERY AREA      | 0.0             | 0.0                 | 0.0                | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| -          | -        | ADULTIP8-DELIVERY AREA      | 0.0             | 0.0                 | 0.0                | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| -          | -        | STROKE-DELIVERY AREA        | 0.0             | 0.0                 | 0.0                | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| -          | -        | ADULTIP9-DELIVERY AREA      | 0.0             | 0.0                 | 0.0                | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| Fraility   | Catering | D1-CATERING                 | 3.61            | 3.61                | 0.0                | 2              | 2                  | 0                 | No failed delivery pressure identified. |
| Fraility   | Linen    | D1-LINEN                    | 3.01            | 3.01                | 0.0                | 1              | 1                  | 0                 | No failed delivery pressure identified. |
| Fraility   | Linen    | D1-LINEN_2                  | 2.99            | 2.99                | 0.0                | 1              | 1                  | 0                 | No failed delivery pressure identified. |
| Fraility   | Pharmacy | D1-PHARMACY                 | 11.97           | 11.97               | 0.0                | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| Fraility   | Stores   | D1-STORES                   | 8.68            | 8.68                | 0.0                | 4              | 4                  | 0                 | No failed delivery pressure identified. |
| Fraility   | Waste    | D1-WASTE                    | 13.45           | 13.45               | 0.0                | 4              | 4                  | 0                 | No failed delivery pressure identified. |
| SDEC       | Waste    | D2-WASTE                    | 13.95           | 13.95               | 0.0                | 4              | 4                  | 0                 | No failed delivery pressure identified. |

| Department      | Category | Location      | Current area m <sup>2</sup> | Recommended area m <sup>2</sup> | Additional area m <sup>2</sup> | Current spaces | Recommended spaces | Additional spaces | Reason                                  |
|-----------------|----------|---------------|-----------------------------|---------------------------------|--------------------------------|----------------|--------------------|-------------------|-----------------------------------------|
| SDEC            | Pharmacy | D2-PHARMACY   | 21.74                       | 21.74                           | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| Urgent Care     | Catering | D3-CATERING   | 4.01                        | 4.01                            | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| Urgent Care     | Linen    | D3-LINEN      | 2.91                        | 2.91                            | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| Urgent Care     | Linen    | D3-LINEN_2    | 2.98                        | 2.98                            | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| Urgent Care     | Catering | D3-CATERING_2 | 3.94                        | 3.94                            | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| Urgent Care     | Pharmacy | D3-PHARMACY   | 11.58                       | 11.58                           | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| Urgent Care     | Stores   | D3-STORES     | 6.63                        | 6.63                            | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| Urgent Care     | Waste    | D3-WASTE      | 15.03                       | 15.03                           | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| Pharmacy        | Waste    | D7-WASTE      | 20.62                       | 20.62                           | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| Pharmacy        | Stores   | D7-STORES     | 48.96                       | 48.96                           | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| Pharmacy        | Pharmacy | D7-PHARMACY   | 0.0                         | 0.0                             | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| Central Imaging | Waste    | D8-WASTE      | 27.71                       | 27.71                           | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| Central Imaging | Linen    | D8-LINEN      | 4.01                        | 4.01                            | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| Central Imaging | Linen    | D8-LINEN_2    | 3.28                        | 3.28                            | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| Central Imaging | Pharmacy | D8-PHARMACY   | 8.62                        | 8.62                            | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |

| Department             | Category | Location     | Current area m <sup>2</sup> | Recommended area m <sup>2</sup> | Additional area m <sup>2</sup> | Current spaces | Recommended spaces | Additional spaces | Reason                                  |
|------------------------|----------|--------------|-----------------------------|---------------------------------|--------------------------------|----------------|--------------------|-------------------|-----------------------------------------|
| Central Imaging        | Stores   | D8-STORES    | 16.6                        | 16.6                            | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| OPD1                   | Waste    | D4-WASTE     | 11.15                       | 11.15                           | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| OPD1                   | Pharmacy | D4-PHARMACY  | 14.73                       | 14.73                           | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| OPD3                   | Waste    | D44-WASTE    | 8.77                        | 8.77                            | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| OPD3                   | Pharmacy | D44-PHARMACY | 12.04                       | 12.04                           | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| OPD4                   | Waste    | D45-WASTE    | 8.59                        | 8.59                            | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| OPD4                   | Pharmacy | D45-PHARMACY | 11.95                       | 11.95                           | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| OPD2                   | Waste    | D5-WASTE     | 11.38                       | 11.38                           | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| OPD2                   | Stores   | D5-STORES    | 10.33                       | 10.33                           | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| OPD2                   | Pharmacy | D5-PHARMACY  | 11.78                       | 11.78                           | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| OPD4                   | Stores   | D45-STORES   | 10.39                       | 10.39                           | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| Mortuary               | Waste    | D6-WASTE     | 13.14                       | 13.14                           | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| Mortuary               | Stores   | D6-STORES    | 26.45                       | 26.45                           | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| Essential Services Lab | Waste    | D15-WASTE    | 23.11                       | 23.11                           | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| Resturant              | Waste    | D9-WASTE     | 11.83                       | 11.83                           | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |

| Department      | Category | Location     | Current area m <sup>2</sup> | Recommended area m <sup>2</sup> | Additional area m <sup>2</sup> | Current spaces | Recommended spaces | Additional spaces | Reason                                  |
|-----------------|----------|--------------|-----------------------------|---------------------------------|--------------------------------|----------------|--------------------|-------------------|-----------------------------------------|
| Resturant       | Stores   | D9-STORES    | 28.27                       | 28.27                           | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| Minor Injuries  | Stores   | D10-STORES   | 14.85                       | 14.85                           | 0.0                            | 5              | 5                  | 0                 | No failed delivery pressure identified. |
| Minor Injuries  | Linen    | D10-LINEN    | 2.95                        | 2.95                            | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| Minor Injuries  | Pharmacy | D10-PHARMACY | 8.0                         | 8.0                             | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| ED-Paediatrics  | Stores   | D13-STORES   | 0.0                         | 0.0                             | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| ED-Paediatrics  | Pharmacy | D13-PHARMACY | 0.0                         | 0.0                             | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| Birthing Suite  | Waste    | D19-WASTE    | 11.92                       | 11.92                           | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| Birthing Suite  | Stores   | D19-STORES   | 33.44                       | 33.44                           | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| Ambulatory Care | Waste    | D17-WASTE    | 11.96                       | 11.96                           | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| Neonatal        | Waste    | D18-WASTE    | 14.64                       | 14.64                           | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| Neonatal        | Linen    | D18-LINEN    | 9.84                        | 9.84                            | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| Neonatal        | Pharmacy | D18-PHARMACY | 14.46                       | 14.46                           | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |
| Birthing Suite  | Pharmacy | D19-PHARMACY | 10.15                       | 10.15                           | 0.0                            | 0              | 0                  | 0                 | No failed delivery pressure identified. |

## Task detail grouped by AMR

### AMR AMR-A-1

| Task                  | Outcome   | From        | To          | Route stops                                                                                                         | Start               | Finish              | Duration | Wait |
|-----------------------|-----------|-------------|-------------|---------------------------------------------------------------------------------------------------------------------|---------------------|---------------------|----------|------|
| GEN_PHARMACY_D1_00001 | completed | D7-PHARMACY | D1-PHARMACY | Pickup: D7-PHARMACY → Drop-off: D4-PHARMACY → Drop-off: D2-PHARMACY → Drop-off: D3-PHARMACY → Drop-off: D1-PHARMACY | 01/01/2026 08:00:00 | 01/01/2026 08:10:32 | 10:32    | 0:00 |
| GEN_PHARMACY_D2_00002 | completed | D7-PHARMACY | D2-PHARMACY | Pickup: D7-PHARMACY → Drop-off: D4-PHARMACY → Drop-off: D2-PHARMACY → Drop-off: D3-PHARMACY → Drop-off: D1-PHARMACY | 01/01/2026 08:00:00 | 01/01/2026 08:10:32 | 10:32    | 0:00 |
| GEN_PHARMACY_D3_00003 | completed | D7-PHARMACY | D3-PHARMACY | Pickup: D7-PHARMACY → Drop-off: D4-PHARMACY → Drop-off: D2-PHARMACY → Drop-off: D3-PHARMACY → Drop-off: D1-PHARMACY | 01/01/2026 08:00:00 | 01/01/2026 08:10:32 | 10:32    | 0:00 |
| GEN_PHARMACY_D4_00004 | completed | D7-PHARMACY | D4-PHARMACY | Pickup: D7-PHARMACY → Drop-off: D4-PHARMACY → Drop-off: D2-PHARMACY → Drop-off: D3-PHARMACY → Drop-off: D1-PHARMACY | 01/01/2026 08:00:00 | 01/01/2026 08:10:32 | 10:32    | 0:00 |
| RETURN-AMR-A-1-1      | completed | D1-PHARMACY | AMR-CENTRE  | D1-PHARMACY → AMR-CENTRE                                                                                            | 01/01/2026 08:10:32 | 01/01/2026 08:13:26 | 2:54     | 0:00 |
| GEN_PHARMACY_D1_00005 | completed | D7-PHARMACY | D1-PHARMACY | Pickup: D7-PHARMACY → Drop-off: D4-PHARMACY → Drop-off: D2-PHARMACY → Drop-off: D3-PHARMACY → Drop-off: D1-PHARMACY | 02/01/2026 08:00:00 | 02/01/2026 08:10:32 | 10:32    | 0:00 |
| GEN_PHARMACY_D2_00006 | completed | D7-PHARMACY | D2-PHARMACY | Pickup: D7-PHARMACY → Drop-off: D4-PHARMACY → Drop-off: D2-PHARMACY → Drop-off: D3-PHARMACY → Drop-off: D1-PHARMACY | 02/01/2026 08:00:00 | 02/01/2026 08:10:32 | 10:32    | 0:00 |
| GEN_PHARMACY_D3_00007 | completed | D7-PHARMACY | D3-PHARMACY | Pickup: D7-PHARMACY → Drop-off: D4-PHARMACY → Drop-off: D2-PHARMACY → Drop-off: D3-PHARMACY → Drop-off: D1-PHARMACY | 02/01/2026 08:00:00 | 02/01/2026 08:10:32 | 10:32    | 0:00 |



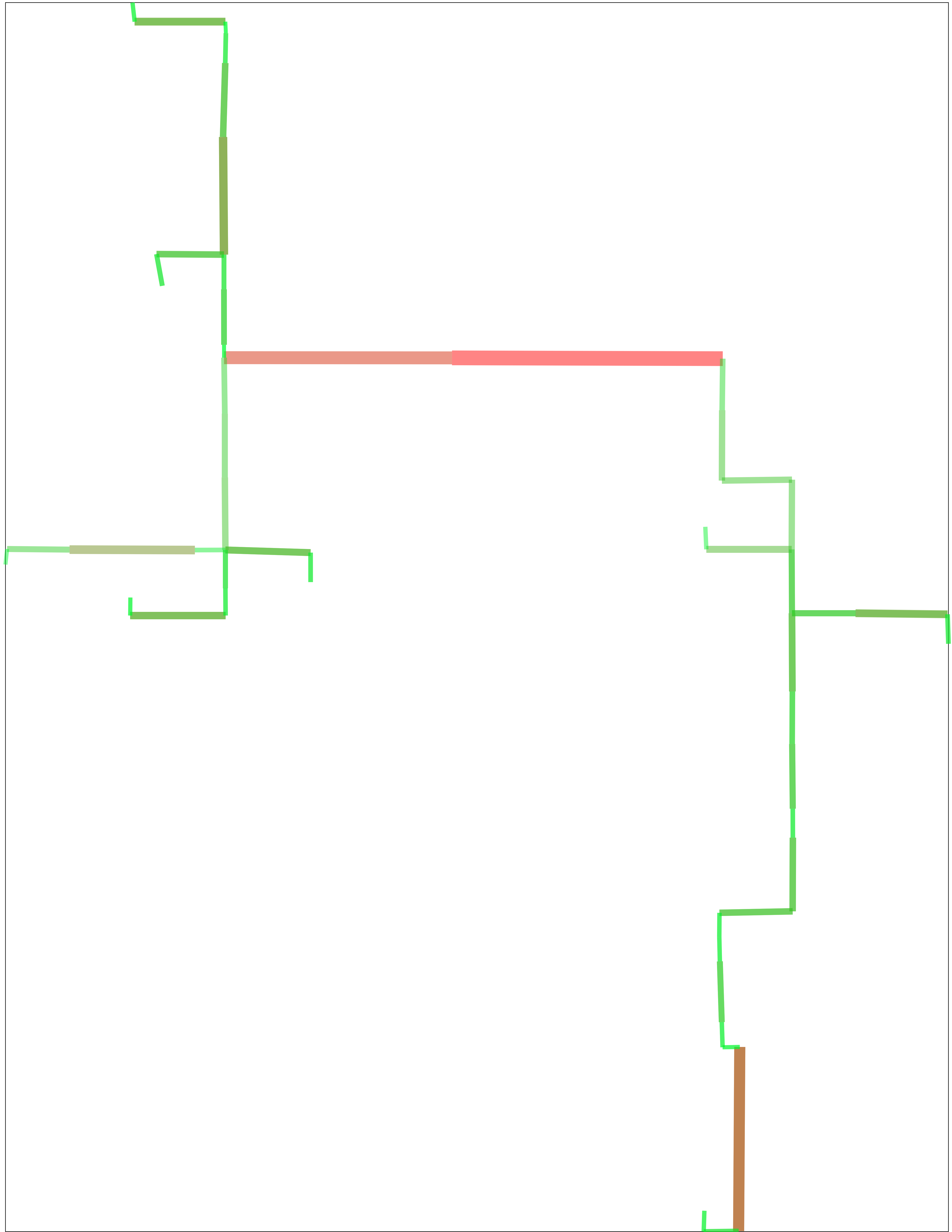
| Task                  | Outcome   | From        | To          | Route stops                                                                                                         | Start               | Finish              | Duration | Wait |
|-----------------------|-----------|-------------|-------------|---------------------------------------------------------------------------------------------------------------------|---------------------|---------------------|----------|------|
| GEN_PHARMACY_D4_00008 | completed | D7-PHARMACY | D4-PHARMACY | Pickup: D7-PHARMACY → Drop-off: D4-PHARMACY → Drop-off: D2-PHARMACY → Drop-off: D3-PHARMACY → Drop-off: D1-PHARMACY | 02/01/2026 08:00:00 | 02/01/2026 08:10:32 | 10:32    | 0:00 |
| RETURN-AMR-A-1-3      | completed | D1-PHARMACY | AMR-CENTRE  | D1-PHARMACY → AMR-CENTRE                                                                                            | 02/01/2026 08:10:32 | 02/01/2026 08:13:30 | 2:58     | 0:00 |

**AMR AMR-B-1**

| Task             | Outcome   | From        | To          | Route stops              | Start               | Finish              | Duration | Wait |
|------------------|-----------|-------------|-------------|--------------------------|---------------------|---------------------|----------|------|
| T1               | completed | KITCHEN     | D1-CATERING | KITCHEN → D1-CATERING    | 01/01/2026 08:06:00 | 01/01/2026 08:11:25 | 5:25     | 0:00 |
| RETURN-AMR-B-1-2 | completed | D1-CATERING | AMR-CENTRE  | D1-CATERING → AMR-CENTRE | 01/01/2026 08:11:25 | 01/01/2026 08:14:30 | 3:05     | 0:00 |



Congestion heatmap - Floor 0





Congestion heatmap - Floor 2

